

EE1101: Circuits and Network Analysis - Quiz 6

Name: _____

Roll No: _____

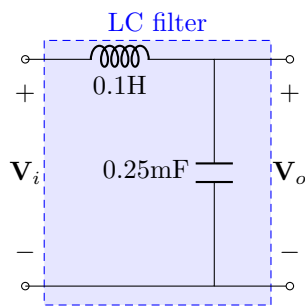
Instructions

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2. Go through all the questions carefully. Any misinterpretations will **not** be considered. Write your solutions in the designated space provided for each question. Answers written outside these areas will not be accepted.
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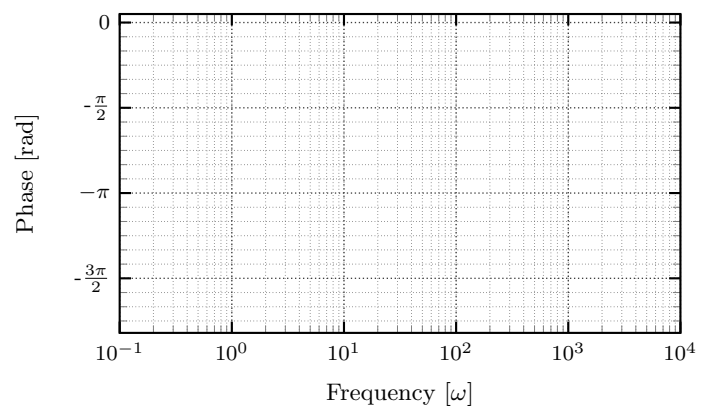
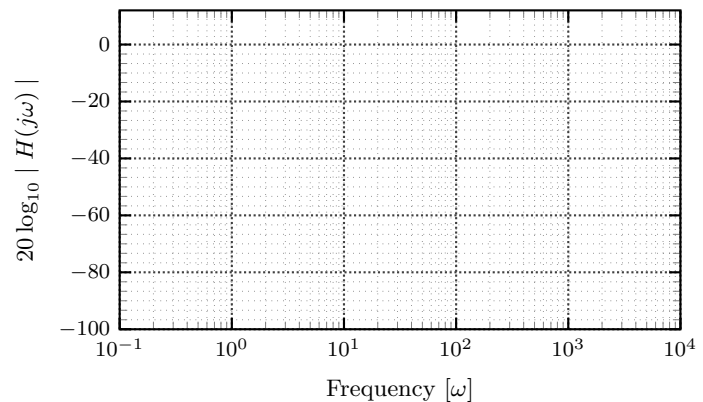
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1. (3 points) The circuit shown in Fig. 1(a) is a LC filter. Let $\mathbf{G}(s)$ denote the transfer function $\mathbf{G}(s) = \frac{\mathbf{V}_o}{\mathbf{V}_i}$. Plot the frequency response of the filter. Use the graph provided adjacent to the circuit.



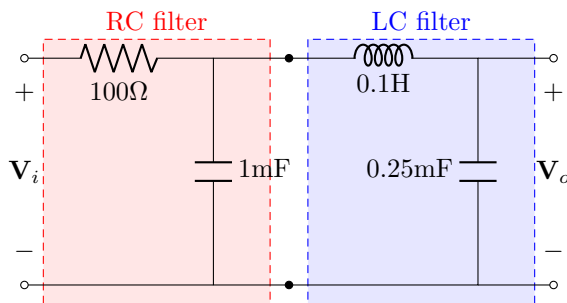
(a) Circuit



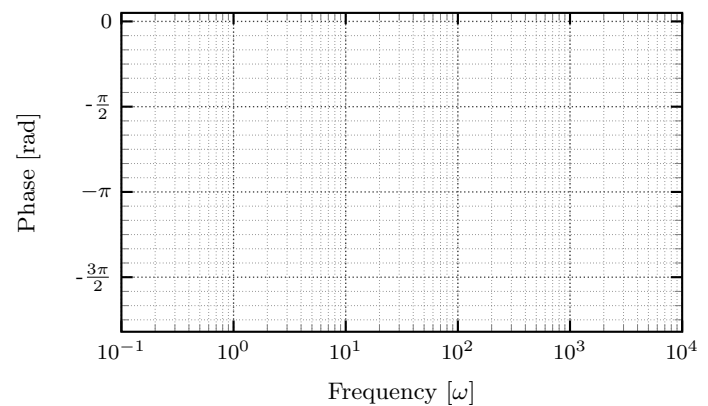
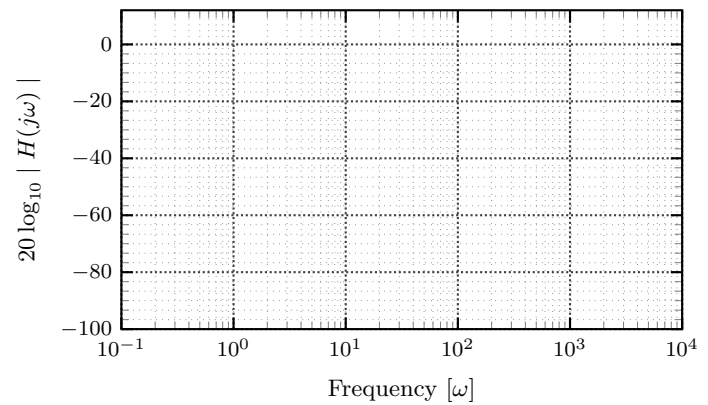
(b) Frequency Response

Fig. 1: For question 1

2. (4 points) Consider a scenario wherein an RC filter and an LC filter are connected in a cascade manner (as shown in Fig. 2). Let $\mathbf{H}(s)$ denote the transfer function $\mathbf{H}(s) = \frac{\mathbf{V}_o}{\mathbf{V}_i}$. Plot the frequency response of the filter. Use the graph provided adjacent to the circuit.



(a) Circuit



(b) Frequency Response

Fig. 2: For question 2

3. (3 points) Determine the frequency at which the circuit shown in Fig. 3 will be under resonance. Enter your answer here: _____

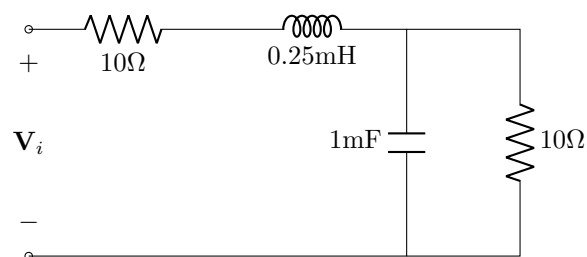


Fig. 3: Circuit for Question 3

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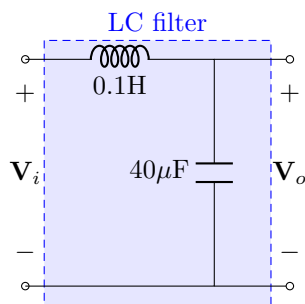
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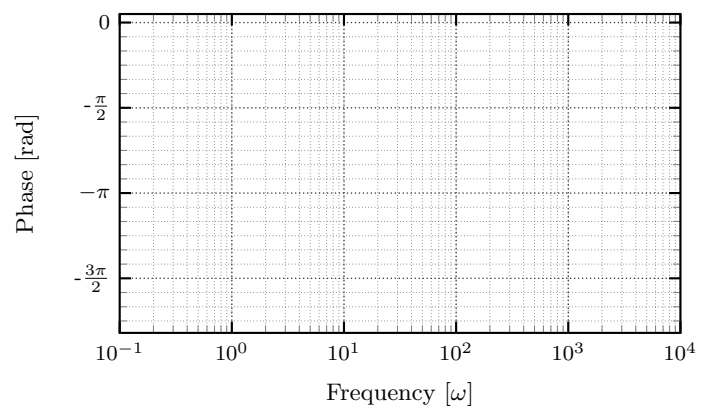
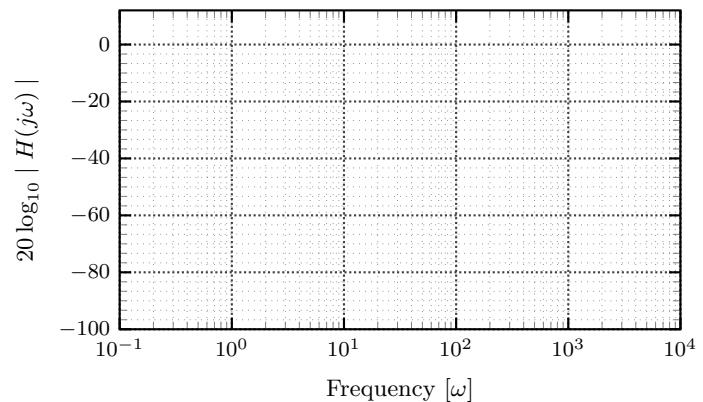
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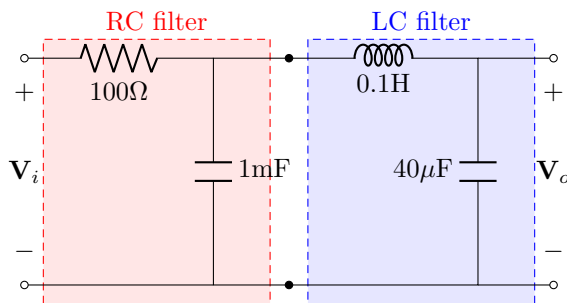
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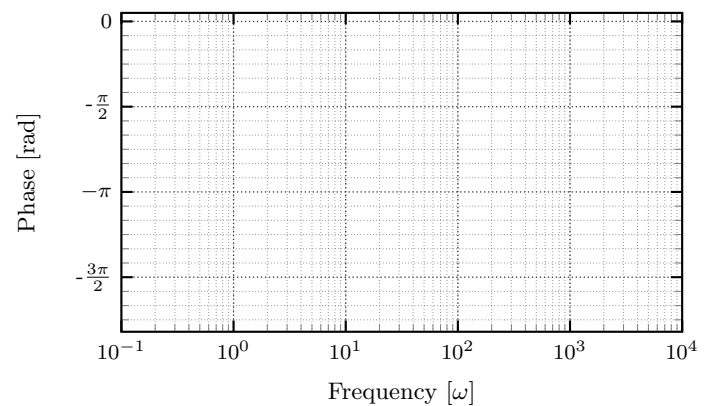
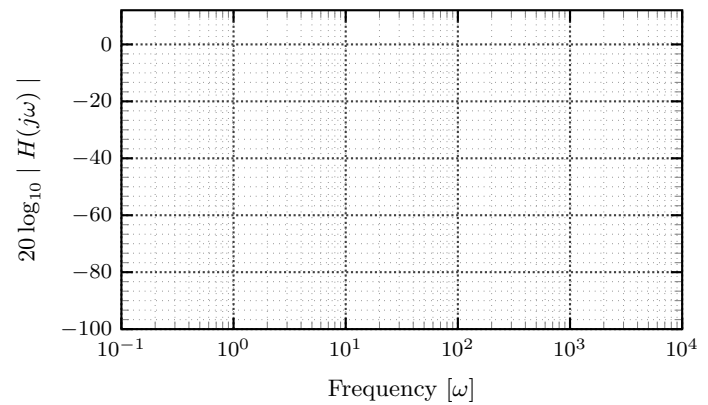
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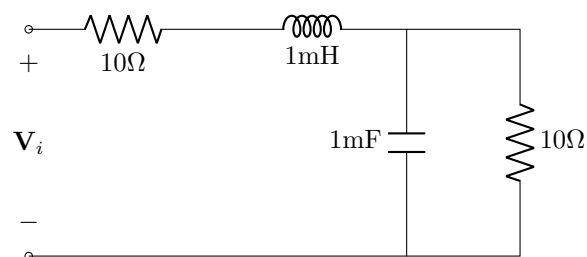


Fig. 3: Circuit for Question 3

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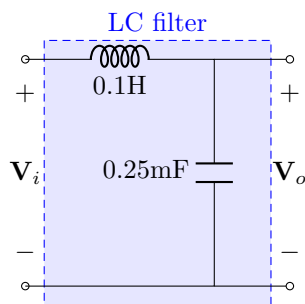
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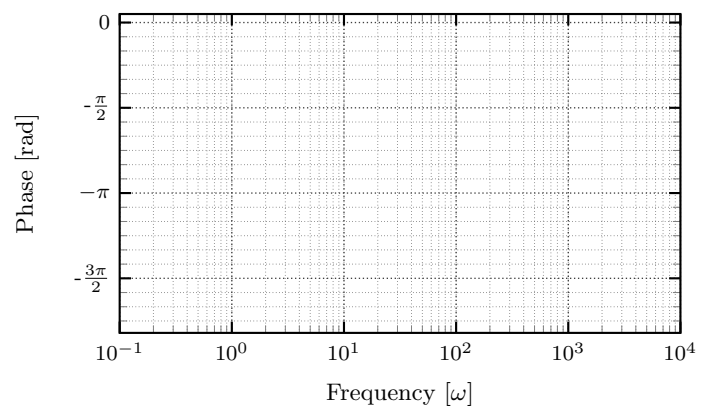
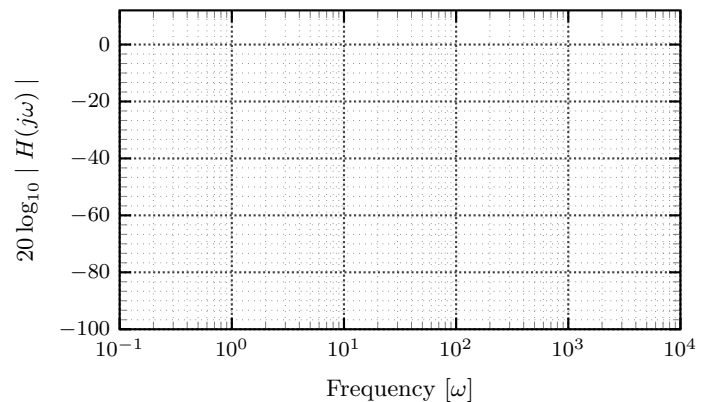
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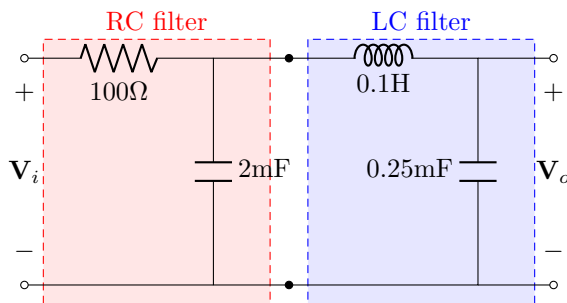
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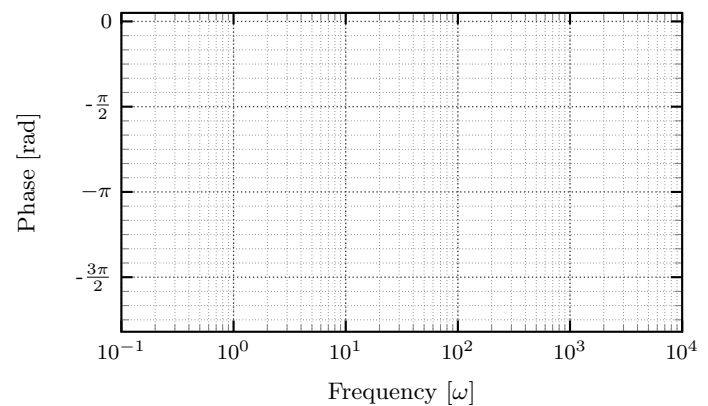
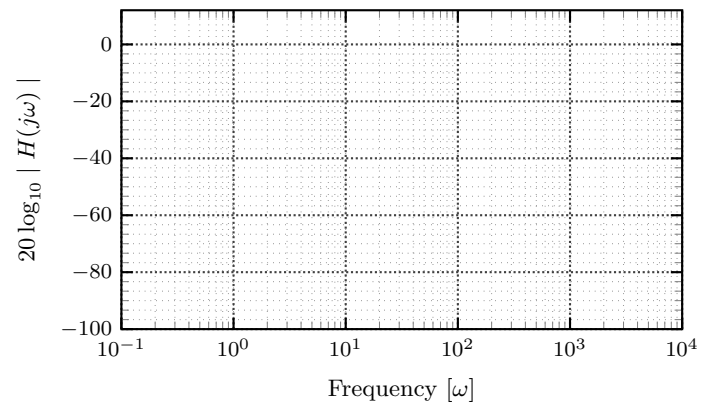
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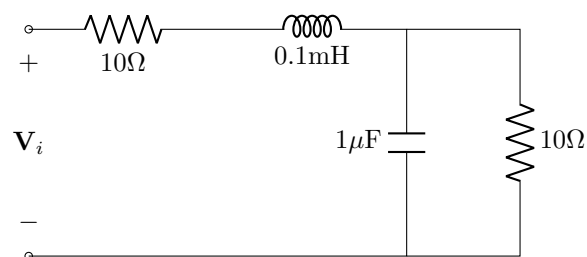


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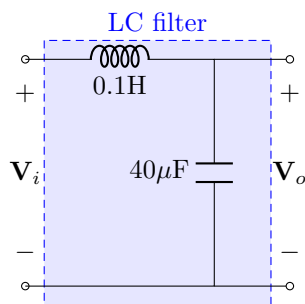
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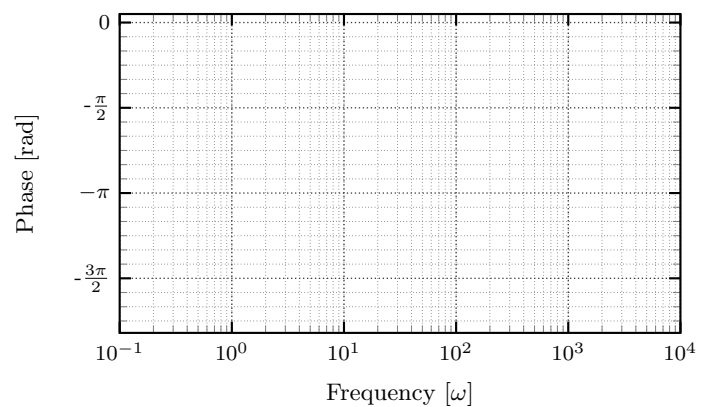
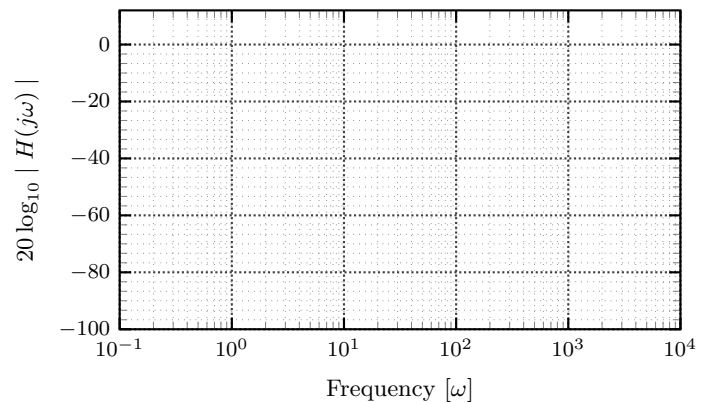
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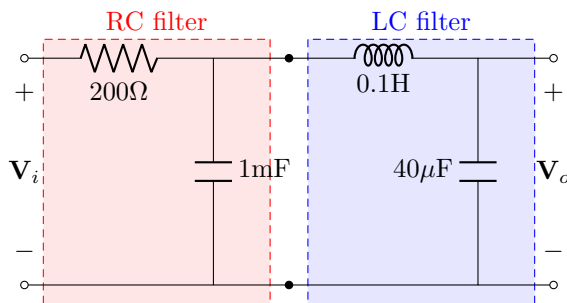
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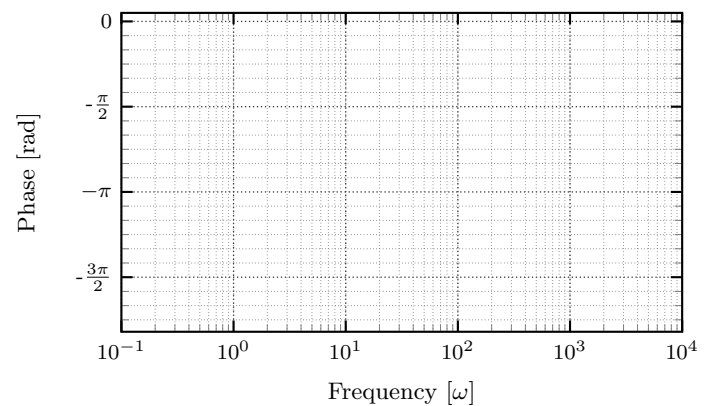
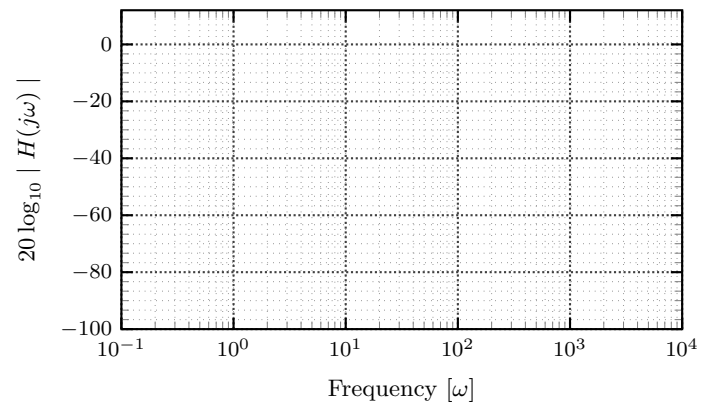
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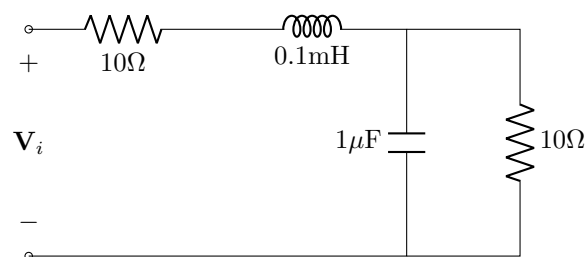


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